

Name: *Nadia Begum*

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Objective

Highly motivated embedded systems engineer with hands-on experience in IoT and microcontroller-based automation. Seeking to apply my technical skills in firmware programming, sensor integration, and hardware interfacing to develop efficient embedded solutions.

Summary of Qualifications

- Bachelor of Engineering in Electronics and Communication from Colorado State University
- Strong background in microcontroller programming (Arduino, ESP32, STM32)
- Skilled in C, C++, Python, and embedded C
- Experience with IoT protocols (MQTT, HTTP, BLE, LoRa)
- Proficient in PCB design, circuit testing, and sensor interfacing
- Excellent troubleshooting and teamwork abilities

Education

Bachelor of Engineering in Electronics and Communication

Colorado State University, Fort Collins, CO

GPA: 3.85 / 4.0

Technical Skills

- **Programming Languages:** C, C++, Embedded C, Python
- **Microcontrollers:** Arduino, ESP32, STM32, Raspberry Pi
- **Communication Protocols:** I2C, SPI, UART, MQTT, BLE
- **Sensors:** DHT11, PIR, Ultrasonic, IR, MPU6050
- **Tools:** Proteus, KiCad, MATLAB, Arduino IDE, PlatformIO
- **Networking:** Wi-Fi, Bluetooth, LoRaWAN
- **OS:** Linux, Windows

Experience

Embedded Systems Engineer | Innovatek Solutions

- Designed firmware for IoT-enabled health monitoring devices using ESP32
- Developed real-time sensor data acquisition and wireless communication modules
- Integrated cloud-based data logging using MQTT protocol
- Tested and calibrated multiple sensors for accuracy and stability

Intern | SmartTech Embedded Labs

- Worked on automation system using Arduino and Raspberry Pi
- Built prototypes for smart home applications involving motion and temperature sensors

Projects

- **Smart Baby Monitoring System:** Developed IoT-based system using ESP32 to monitor baby's temperature and movement, sending alerts via WhatsApp using IFTTT.
- **IoT Weather Station:** Designed and coded weather monitoring system using DHT11 and BMP180 sensors, integrated with ThingSpeak cloud.

Certifications

- Embedded Systems Design Certification (Coursera)
- IoT Fundamentals by Cisco Networking Academy

Languages

- English (Fluent)
- German (Beginner)